

torch.addcddiv#

<https://pytorch.org/docs/stable/generated/torch.addcddiv.html>

Description:

Performs the element-wise division of `tensor1` by `tensor2`, multiplies the result by the scalar value and adds it to input. Integer division with `addcddiv` is no longer supported, and in a future release `addcddiv` will perform a true division of `tensor1` and `tensor2`. The historic `addcddiv` behavior can be implemented as $(\text{input} + \text{value} * \text{torch.trunc}(\text{tensor1} / \text{tensor2})).\text{to}(\text{input.dtype})$ for integer inputs and as $(\text{input} + \text{value} * \text{tensor1} / \text{tensor2})$ for float inputs. The future `addcddiv` behavior is just the latter implementation: $(\text{input} + \text{value} * \text{tensor1} / \text{tensor2})$, for all dtypes. The shapes of `input`, `tensor1`, and `tensor2` must be broadcastable. For inputs of type `FloatTensor` or `DoubleTensor`, `value` must be a real number, otherwise an integer.

Function Signature

```
torch.addcddiv(input, tensor1, tensor2, *, value=1, out=None)  
→ Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

value (Number, optional) – multiplier for $\text{tensor1}/\text{tensor2}$ / $\text{tensor1}/\text{tensor2}$ out (Tensor, optional) – the output tensor.

Code Examples

```
>>> t = torch.randn(1, 3)  
>>> t1 = torch.randn(3, 1)  
>>> t2 = torch.randn(1, 3)  
>>> torch.addcddiv(t, t1, t2, value=0.1)  
tensor([[[-0.2312, -3.6496,  0.1312],  
          [-1.0428,  3.4292, -0.1030],  
          [-0.5369, -0.9829,  0.0430]]])
```

Notes & Warnings

WARNING

Warning Integer division with `addcdiv` is no longer supported, and in a future release `addcdiv` will perform a true division of `tensor1` and `tensor2`. The historic `addcdiv` behavior can be implemented as $(\text{input} + \text{value} * \text{torch.trunc}(\text{tensor1} / \text{tensor2})).\text{to}(\text{input.dtype})$ for integer inputs and as $(\text{input} + \text{value} * \text{tensor1} / \text{tensor2})$ for float inputs. The future `addcdiv` behavior is just the latter implementation: $(\text{input} + \text{value} * \text{tensor1} / \text{tensor2})$, for all dtypes.

Detailed Documentation

Documentation

Performs the element-wise division of `tensor1` by `tensor2`, multiplies the result by the scalar value and adds it to input.

Integer division with `addcdiv` is no longer supported, and in a future release `addcdiv` will perform a true division of `tensor1` and `tensor2`. The historic `addcdiv` behavior can be implemented as `(input + value * torch.trunc(tensor1 / tensor2)).to(input.dtype)` for integer inputs and as `(input + value * tensor1 / tensor2)` for float inputs. The future `addcdiv` behavior is just the latter implementation: `(input + value * tensor1 / tensor2)`, for all dtypes.

The shapes of `input`, `tensor1`, and `tensor2` must be broadcastable.

For inputs of type `FloatTensor` or `DoubleTensor`, `value` must be a real number, otherwise an integer.

`input` (Tensor) – the tensor to be added

`tensor1` (Tensor) – the numerator tensor

`tensor2` (Tensor) – the denominator tensor

`value` (Number, optional) – multiplier for $\frac{\text{tensor1}}{\text{tensor2}}$

`out` (Tensor, optional) – the output tensor.

